

Carbon

Carbon is essential to all life on Earth. It is in the atmosphere, oceans, plants, soil, rocks, and even our bodies. The total amount of carbon in the world never increases or decreases, but it is always moving around and changing form. This process is called the carbon cycle.

In the atmosphere

In the air, carbon is combined with oxygen in carbon dioxide (CO_2). CO_2 is known as a greenhouse gas because it traps heat. Reducing the amount of CO_2 in the atmosphere helps prevent global warming.

Absorbing

Plants use light energy from the sun to combine carbon dioxide from the air with water to make food. This is called photosynthesis.

Plants help lower the amount of carbon dioxide in the atmosphere.

Dead animals and plants release CO_2 when they decay.

Carbon can stay trapped at the bottom of deep oceans for thousands, or even millions, of years.

Storing

Over millions of years, dead organisms, which contain carbon, turn into fossil fuels such as coal and oil.

Rain forests Trees absorb lots of CO_2 , helping reduce greenhouse gas in the atmosphere. However, deforestation (cutting down lots of trees) gets in the way of this natural process.



Rocks Most of Earth's carbon—about 64 trillion tons—is stored in rocks. However, wind, rain, and ice can break down rock and release CO_2 back into the ocean or atmosphere.

